

Lec 32 GAN Architecture

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Lecture	32 / 63 + intro
Duration	50:13
Captions	English (auto-generated)
Words	7,533
Video	uWm5rUjpZ_4
Playlist	Fundamentals of Generative AI and LLMs

Auto-generated YouTube captions, lightly cleaned (removed [music] tags, grouped into paragraphs). These are **verbatim lecture captions**, not edited notes.

Transcript

[00:00] Hello everyone, welcome to the lecture series on fundamentals of generative AI and large language models. We are in the week five. In the previous session, we had seen what are the limitations of variational autoencoders and we have seen what is the key motivation for GAN and towards the end of that particular uh session we have also seen something called as what exactly is adversarial learning. We did not go in detailed about uh the generator and discriminator. In this particular session uh in today's session we will look into v uh g architecture where we will look into in detail what exactly is a generator and what is a discriminator.

[00:43] So this is a generator uh uh training. So let us uh see what exactly it is uh and what we are looking into now is let us understand how a generator is trained in a GAN. Now um the the process uh let me first okay the process starts um or the process begins with a random noise vector z and if you look into z is having some dimension here this means z is a d -dimensional vector and its values are usually sampled from an distribution and that distribution here is an standard normal distribution. So the very first step generator is the one which has to generate the fake samples from where it will generate the fake samples. It has to generate from some random noise and that random noise is being sampled from an Gaussian

[01:41] distribution. That is what is the first step. Now uh um whatever is this random noise z uh does not initially contain any meaningful uh information. uh it um it is uh simply a set of some random numbers. Okay. Next, this random noise vector is passed as an input to a generator and generator here is a neural network. Whenever you say a neural network, a neural network will have certain parameters and that parameter we are denoting it as ϕ and here ϕ represents the trainable parameters of a generator. Now for this generator um what is that we are passing? We are passing z . So when you pass z then generator going uh will transform this lowdimensional latent vector into an generated image. So generator is the one

[02:40] which will take z as an input and then it is going to convert that into it is going to transform this lowdimensional random vector into an generated image. So generated image here is denoted as this which is nothing but $\hat{x}$. Okay. I hope this bottom part is clear. Random noise which is sampled from gaussian distribution is passed as an input to a generator. Generator is a neural network which will transform this noise into an generated image. Here the generated image is denoted as this $G(z)$ or it is also $\hat{x}$. Right? Now uh once we get this generated image now this generated image has initially this this can have any number of u the dimension is not fixed here but by the time you generate

[03:39] an image this generated image has the same dimension as the real images in the training uh data set. So the size should be same as the real image. Why? Why is that we have to do? Because we are passing it to a discriminator and discriminator will also see a real images. Therefore, generated image size should be same as the real images. For example, if the real image contains a grayscale image and if that image is of size 28×28 , then the generator also produces an image of size 28×28 .

[04:18] Suppose if an real image is a colored one. If the data set contains an color image which is having a size $64 \times 64 \times 3$ uh then the generated image also will have the same size. Why this is necessary? Because uh both real and generated image must be compatible with the same discriminator network because discriminator is the one which will be seeing real image as well as generated image. If generated image size is very less is it not discriminator can easily identify that this is an generated one. So in order to avoid that first aim first motive here is dis generator has to generate an image and the size has to match with the discriminator.

[05:09] Okay. So far uh let me once again revise to you because this is very very important architecture and your entire week six is once again based on the week five. So first you have an random noise. Random noise is sampled from a gaussian distribution. That means whatever the values that you are taking these are the values which are sampled. This is a um you know hyperparameter. If you fix a latent size as 100 then you will take 100 such values from this gaussian distribution and then pass that 100 values to a generator. Generator here is a neural network. When you say neural network you will have a parameters.

[05:52] Those parameters here are five. Uh these are learnable trainable parameters. This generator takes Z and transforms this low level into an generated image. So this generated image is uh to this generator you are passing Z and you are getting an generated image and you are calling it as $\hat{X}$. This generated image size should be same as the uh data set original data set size. Now you can ask how will we get to know? So when you are coding you will explicitly specify that towards the end you should get this particular size. Okay why is that we have to ensure it is same size because ultimately the generated image will be sent to an discriminator. So when a discriminator when you send it to a discriminator if the size is not

[06:44] matching it's a direct indication to an discriminator right that it's an generated image. Therefore, it will ensure that uh the size will remain the same. And second, one more reason why size has to be same is discriminator will see original images also. Original image will be of one size. Generated image if it is of different size. How is that the network you have to design? Correct. So because of that both images should be of same size. Okay. I think up till here it is clear. Now whatever the images the um that the generator has generated which is this is now passed to an discriminator. So here the generated image is then passed to an discriminator. Discriminator is a neural network. If discriminator is a

[07:35] neural network then that neural network will have certain parameters and that parameters here is θ that can be weights and bias. So what is that input you are passing to a discriminator? Uh the input that you are passing to a discriminator is this. Um so um uh generated image is this that is being passed to an discriminator and um here um um you know θ represents the parameters. Now what discriminator will do?

[08:06] discriminator will examine the generated image and say whether it is real or fake. Now how is that it will u give us how it will tell whether it's real or fake? It is going to give us one probability and that probability value will be always between zero and one. Okay. So which means the discriminator is going to give us $\hat{y}$. What is $\hat{y}$? Here is the predicted value. What is that predicted value? Discriminator should tell whether it is a generated image or not. Discriminator has to tell that. So $\hat{y}$ is the output that the discriminator is going to give us. So that $\hat{y}$ is nothing but to this input the generator uh is been pass generated input is being passed to discriminator and discriminator is going to give us

[09:00] the um probability. Now suppose if the probability is 0.2. Suppose if the discriminator has given this probability. What is the meaning of this? This means that the discriminator assigns only 20% probability that the generated image is real. In other words, discriminator is it not discriminator is very confident that um image is generated or it is a fake image. Right? So this is how um up till this particular uh step uh I hope up till here it is clear to you uh generate u generated image is being passed as an input to an discriminator and the discriminator will take that generated

one as an input and it is going to give us the probability. Okay. Now when it has predicted $\hat{y}$ the next step is it has to calculate the

[09:59] loss right. What is the loss here? Generator loss uh which is denoted as L_G . Now if uh if you have to calculate the loss, you need to have the ground truth also. You need to if $\hat{y}$ is the output, you need to have a Y also, right? So during a generator training, what we are discussing is the generator training. What is the ultimate aim of a generator? The generator wants the discriminator to classify the generated image as real. That's the ultimate aim.

[10:33] U whatever generator is generating an image, discriminator should classify it as uh real. If it has to classify it as real, what should be uh the ground truth, the desired y value y is one, $y = 1$. Just think uh carefully. Now if a generator uh is generating an image and for that image if the discriminator is always giving a very low probability which means discriminator is easily classifying uh the image as generated one. But ultimately what is the role of a generator? it has to improve and then provide more realistic image. If it has to improve its ground truth should be always one. If the ground truth is one then uh if initially if it is uh giving the probability as 0.2

[11:37] when you compare it with the ground truth one the loss is too much. When the loss is too much is it not very good information is coming to an generator. If the ground truth is zero suppose if the ground truth is zero because if you are thinking that it is generated image so y should be zero then is it not this error will be very less when you compare with zero when you compare it with zero if the error is very less ultimately what is that you are informing to an generator that whatever image you are generating is good then where is that realistic image you are getting I hope this is clear right so that is the reason why y is assumed assumed to be one. So uh this target is deliberately set to one because the objective of a

[12:26] generator is to fool the discriminator and it has to generate more realistic image. So I hope uh up till this point it is uh clear to you. So this is important. Um I'm before I go to the next slide I'm once again quickly revising. So there is an loss uh uh sorry there is an uh noise. This noise is passed to an generator. Generator will generate an image. This generated image is passed to an discriminator and the discriminator is going to give us the y probability and this probability y is a predicted value $\hat{y}$. This $\hat{y}$ should be compared with the ground truth and that ground truth is one. Then you are going to calculate the loss. Now uh once this flow is clear, let me explain you the terms that are used here. So uh

[13:19] this is G generator parameter generator network. The network will have parameter. So G . Now for this what is that you are passing? You are passing a Z . What is Z noise that is passed as an input and what is that you get from this? You get the y image generated image. generated image is denoted as this. This is how you

are denoting the generated image. Why? This is the one which is generating and it is generating with the help of z . So this is the generated image and we in a short we are representing it as $\hat{x}$. Now you are passing it to a discriminator.

[14:01] Discriminator it is a neural network which is having the parameters θ . Now for this what input you are passing? You are passing the generated image. What is generated image? Is it not? This is the generated image. This is the generated image that is passed as an input. So uh when when you pass this uh uh as an input to an discriminator, discriminator is going to give us the prediction. So prediction is on this right. So that predicted value is $\hat{y}$.

[14:38] $\hat{y}$ that is compared with the ground truth. Why ground truth is one that we have discussed already. Ultimately generator has to generate in a realistic image. For the realistic image y is taken as one. Once this is done you are computing the loss. So loss is between $\hat{y}$ and y . So once you compute loss what is the next step? You have to back propagate. And uh when you back propagate is it not you back propagate like this.

[15:08] the ultimately you will reach to an generator. So rather than showing an arrow mark like this, I'm showing a blue dotted arrow marks on the top. So here during back propagation what exactly is happening? Although you come through a discriminator, discriminator has the parameters θ but you are not going to do any updations here. uh gradients will pass through discriminator but θ is not being updated which means discriminator parameters are not being updated. Here the aim is generator. So discriminator will be made will be freezed there. Then only generator parameters are updated. What are parameters? It is fine. So I hope uh generator flow is clear to you. This is important. Second uh next let us look

[15:56] into what exactly is the generator loss. So um go back and then think of binary cross entropy. So is it not binary cross entropy loss here. So now from this let us see how we will derive it to an generator loss. This is a simple binary cross entropy which we are seeing from many um stages. We have seen this before also. Now here if you look into what is our ultimate aim. The aim here is during a generator uh training the generator wants a discriminator's prediction for the generated image to become closer to one. So which means whatever the images that the generator has generated for that when you pass it to a discriminator discriminator uh prediction should be closer to one. That means if the

[16:50] prediction is closer to one, what is the idle meaning? Generator has generated a realistic image. That's the meaning of it. Yes. So that's what is the ultimate aim. During a generator training, the generator wants discriminator prediction which means $\hat{y}$ should be closer to one. Okay. If um so that you know makes the discriminator classify all the generated images as real. Now what is the target y ground truth y that we are considering is it not y is one that we have seen some time back.

Now you are clear why y is 1. Now just substitute y in the given formula in the above formula when y is 1 then $1 - 1$ becomes 0 this entire term becomes zero. Then is it not when y is 1 this term becomes zero. Then you are left out with 1 into

[17:46] $\log \hat{y}$ is minus $\log y$. Is it not? We are obtained with this. Left with this. Now what is $\hat{y}$? $\hat{y}$ is a predicted value. Prediction from where? Prediction from discriminator. What is the input that you have passed? So that discriminator is a predicting generated image. So is it not this is your $\hat{y}$. So substitute that $\hat{y}$ into this um LG that is loss at an uh generator. So is it not this is the formula that you get. This is important concept. So generator loss is this. This is the generator loss. I hope this is clear to you how you have obtained. So you have obtained this because in our loss binary cross entropy we are seeing that y is for a generator training y is one when y is one substitute the formula value into

[18:47] the given loss function. So when the second term becomes zero when the second term becomes zero is it not we are left with only first term and what is $\hat{y}$? $\hat{y}$ is the predicted value from a discriminator. How is that you are representing it is uh discriminator by passing the generated data as the input. So therefore loss function is this. I hope this is clear to you generator loss. So uh what is the entire forward pass? We have a z . z is passed to an generator and the generator is going to produce the generated image this and that is passed to an discriminator um d and d is going to give us the prediction $\hat{y}$ and that uh later you are computing the loss. What is the

[19:38] loss here? The loss is this. This is a loss function of an generator. Now let us say one example. Suppose if model has predicted $\hat{y}$ as 0.2. So if uh 0.2 ultimate aim is it should be closer to one. But what is that the model has predicted? The model has predicted it as 0.2. So substitute the value in this minus $\log 0.2$. Is it not? We are getting this. Is it not the loss is too much? Loss is too high. When generated loss is high what is the meaning? Two meanings. discriminator is very successful in saying that it is a fake image which means a generator has not generated a realistic image. Is it not that these two informations is getting conveyed? Now if this is the loss during a back propagation is it not

[20:28] this information is conveyed to an generator when this information is conveyed is it not generator will improve further. So generator loss is high because the discriminator confidently identifies generated image as a fake. So because of this generator will get the information that I have not generated image properly. So generator will improve. Next now let us see what exactly is the back propagation. So if you recall from our generator uh training we have u you know the overall ultimately when you're back propagating you should back propagate it till the generator parameters you have to update.

[21:09] So what is the overall flow? Is it not? Um you know from loss uh loss you

have obtained from $\hat{x}$ uh $\hat{y}$ that is a predicted value and uh that $\hat{y}$ that you have got from the generated uh data u image that generated we are calling it as $\hat{x}$ and from there you are reaching it to an generator parameters five right this is the flow uh during a back propagation you are calculating the loss from loss you are reaching it to $\hat{y}$ from $\hat{y}$ you are getting it to $\hat{x}$ from $\hat{x}$ to the parameters ϕ . So is it not from loss you first reach to an discriminator from discriminator you will reach to an generator parameters five [snorts] in this there are two important what is $\hat{y}$ what is this I by now you should be

[22:05] very clear what is this notation and $\hat{x}$ is an generated image right now just substitute the values now you are back propagating it back propagating is always and chain rule. Using a chain rule, can I write derivative of loss with respect to five is first you will reach to this generative of derivative of loss with respect to $\hat{y}$. From $\hat{y}$ you will reach to $\hat{x}$ derivative of $\hat{y}$ with respect to $\hat{x}$ and from $\hat{x}$ you will reach to the parameters five using the chain rule. Can we write in this way? Yes. Now uh at the right side you have the values you have what is $\hat{y}$? You have what is $\hat{x}$? Can you just substitute those values into this formula? In this $\hat{y}$, can you bring this value? Can in at this $\hat{x}$ can you

[22:55] bring this value? Is it not? It looks something like this. Substituting the definitions derivative of loss with respect to ϕ . Is it not? You have reached to this point. So all of these are easily differentiable. So therefore we have got uh this value. Now you have to finally update the generator parameters. So what is uh that you have to update those parameters five. So five new is five old minus learning rate derivative of loss with respect to the parameters. So is it not that this? So in this learning the this is the learning rate. Now is it not you have easily updated the parameters during the back propagation. you were easily uh able to update the parameters. Now in this formula you look very carefully we

[23:49] have even θ also uh we have discriminator θ . So during a back propagation during this particular step or uh when you are uh training the generator when you are back propagating although there is a discriminator θ θ is frozen which means there is nothing that you will be doing here. So only the five is updated that is generator parameters have been updated. Now by updating these parameters what is your ultimate aim? What is the purpose of uh uh update? The purpose of this update is to modify the generator. Why you want to modify the generator? So that a discriminator assigns the high probability u if it has if it has assigned high probability then is it not generator is very good then

[24:41] discriminator is not able to identify which is real or which is a fake one. So uh the purpose of this update is to modify the generator so that discriminator assigns the high probability of being real to a generated image. So which means this value should be closer to one. When this value is closer to one uh there are two meanings again discriminator is not able to identify whether it is real or fake. If discriminator

is not able to identify which means the generated images are very much closer to the original one. So the discriminator is definitely used to compute the gradients but its parameters remain fixed during the generator update. So this is the overall meaning of and overall working of a generator.

[25:30] So during this particular phase only generator parameters have been updated. How are generator parameters updated? Is it not we are getting loss from that loss you are um you know going back to a generator and you're updating the parameters in this process discriminator parameters remain fixed. Okay. So with this we have uh finished the generator part. Now let us look into what exactly is a discriminator. Now um so far we have clearly understood the generator its loss function and its back propagation all of that has been made clear. Now uh let us look into a discriminator training. So uh the second network the second network here is discriminator.

[26:20] When you look into there are um uh you know discriminator is the one here which will receive two types of inputs here. It is receiving one from here, one from here. There are two types of inputs. One are the real um images. [snorts] Second one is the generated image. So um and what is the task? The task is to correctly distinguish between them. So now we are only discussing about the uh discriminator. At the discriminator, let us look into the first path and the first path here is real image branch. We are looking into this part. Now we are looking into this.

[26:59] So real image branch. Okay. Um we first take a real image X from the training data set. Let us assume this is a training data set. We take real image X from the training data set. Mathematically we write it as this way. X is sampled from this particular uh distribution. What is this P data of X ? It is the um underlying distribution of the original uh data set or the real data set. Don't get confused just because we have written this we are not um learning any of this probability.

[27:35] No, it is X is been a sample. X is taken from this particular distribution. Okay, this means X is a sample which is drawn from a real data distribution. That's the meaning of this particular line. In practice, we simply select a real image or a mini batch of real images from an available training data set. We simply take this uh data and these real images we are denoting it as X that is passed to a discriminator.

[28:07] What is discriminator? Discriminator is this. So when you pass it to a discriminator, discriminator is going to give us the probability for this particular one. So when it gives the probability what should be the probability what is the role of a discriminator it should clearly tell it is real and it should clearly tell it is a fake. So which means when a real image X is passed to a discriminator discriminator should assign very high probability to uh $d_{\theta}(x)$ right. So can I say that is um prediction made for real image right? So that is what is y_{real} which is uh the $d_{\theta}(x)$ and this

value should be very high. This value represents discriminators estimated probability that the input

[29:03] image is real. Now uh when uh now look at this when yhat is the prediction made for a real image what should be its ground truth what should be y real the y real uh the desired target is y real should be one why y real should be one because it's a real image right when it is a real image the prediction made by the model should be very high and its ground to truth should be one I hope this is clear therefore for a real image the discriminator should produce this $d_{\theta}(x)$ a value which is closer to one. Um I hope this is clear.

[29:43] Suppose I have taken one real image x and that real image x is passed to an discriminator and discriminator is giving the probability 0.9 which means discriminator assigns 90% probability that the image is real. So which means it's a good prediction right. So um okay let us assume we have reached up till this uh point. Now um uh is this u steps clear? You have taken an a real image that real image is passed to a discriminator. Discriminator is giving us the prediction and that prediction we call it as yre yhat real. Uh and that prediction should be closer to one. When that prediction is closer to one which means for a real image the model should assign high probability. If that is the case, what should be its ground truth?

[30:33] Its ground truth should be equivalent to one. Okay. Now let us try to understand what exactly is the second path. Second path is about the generated one. So uh sometime back we've already seen in detail what exactly. So I'll be little fast here. We sample uh z uh that is a random noise vector u and that is sampled from an prior distribution that is gaussian distribution. This noise vector is passed through an generator.

[31:02] So it is uh uh this and it produces an generated image. So can we call it as generated image fake which is nothing but this. Okay. Now the generated image has the same dimension as the real image. So it can be passed through a same discriminator because discriminator you can you don't have two discriminators. It's only one discriminator. Discriminator is being designed based on the real image dimension. Now if you have different dimension then how can you pass it to that same discriminator? Therefore the generated image has the same dimension as the real image so that it can be passed through a same discriminator.

[31:45] Okay. Now the generated image this is what is the generated image that is given to an discriminator d_{θ} . uh it is uh given to. So what is that discriminator will now do? Discriminator produces another probability. What is that another probability? Y fake and hat yhat fake. This value is again interpreted as the probability that the input image is real. Right? Now uh um here this is the the im prediction made by the model. Now if this uh what what should be this value? This value should be closer to zero which means the value will be uh should be less. Now if this value

should be less what should be its correct target or what should be its ground truth? The ground truth should be equivalent to zero. Now don't get

[32:41] confused. Earlier we said it as one but that is for a generator but here it is a discriminator. For a discriminator when you are passing the generated images for a discriminator when you pass the generated image then its uh target value should be closer to zero. Therefore y fake value here is closer to zero. Now if the y fake value is closer to zero which means the uh discriminator should therefore produce this value uh closer to zero. For example uh suppose if the value given by uh this is 0.2 which means the discriminator assigns only 20% probability that the generated image is real. If the generated image is real, if it is assigning only 20%.

[33:40] Which means it has correctly considered that the image is fake image. Right? Okay. This is all with respect to we have reached to this step. Uh I'm repeating the um second branch. First you are taking the noise. Noise is passed to an generator. Generator is generating an image and that generated image is passed to an discriminator and discriminator is taking the generated image and it should assign the probability because it is a generated image for a discriminator this value should be low. If this value is low it should be closer to zero. Therefore its target Y should be zero. Right? So this is for fake and this is for fake. For a real image, let us take real image X.

[34:33] When that is passed to an discriminator and discriminator is has to give us the prediction and this prediction should be high and should be closer to one. Why closer to one? It's a real image. If that is the case, it's a ground truth should be closer to one. And this is all for real image. I hope these two are clear to us. Once this is done, you have to now calculate compute the loss. We are computing the loss here. Uh for the first path and for the generated path, we are computing the loss. Once you compute the loss, you will combine both these two losses. LD that is loss at a discriminator will be a real loss and a fake loss. Then you will back propagate it. when you are back propagating till where you will

[35:24] back propagate you will back propagate only till discriminator and you will update those parameters. Now I hope up till here it is clear to you. Let us now discuss in detail about the loss. Look at this again we are considering an binary cross entropy which is $y \log \hat{y} + 1 - y \log 1 - \hat{y}$. This is a binary cross entropy loss. So once you take this uh you remember there are two things one is real other one is fake. So uh let us now look at the real one. When it is at the real what should be y real the ground truth should be equivalent to one. If it is equivalent to one just go back to the formula and then substitute when this is one 1 minus one becomes zero. So you will be uh when this becomes uh zero this entire term is not

[36:18] there. When this entire term is not there, we are left out with minus 1 into $\log \hat{y}$ we are left with only that right now when we are when we are only left with this we have $\hat{y}$ what is $\hat{y}$ what is a predicted value what is this predicted is it

not you are passing the original image so the $d_u \theta$ of x so it is minus log of x this is what is the loss and the loss at what loss for a real images. So sometime back we have seen uh we are now calculating with this part. Uh so this is the real L_{real} is what we are calculating. So uh I hope this is clear to you how we got the loss term for the real images. y_{real} whatever the model has predicted predicted is $d_u \theta$ of x and for original y_{real} is 1. $\hat{y}$ is d

[37:21] θ of x y_{real} is 1. Just to go back to the formula and substitute when y is 1 then $1 - 1$ becomes 0. This entire term becomes zero. Then you are left out with only this term. You're left out with this term. So what is $\hat{y}$? $\hat{y}$ is uh $d_u \theta$ of x and this is for real images. So y_{real} uh so and what you have obtained is L_{real} um loss. This is a loss uh function uh loss formula what we have obtained this is the first second you have generated also how is that you will be writing for generated. So go back for a generated what should be um the original y fake one the original y should be zero. So when this is zero if you consider this term this term will not exist because y is 0 0 into something becomes and $1 - 0$ will be

[38:21] $1 - \hat{y}$ right now what is $\hat{y}$? $\hat{y}$ is uh the prediction made by the discriminator for a generated images. Prediction made by the discriminator for a generated image. Right? So this is $\hat{y}$ and it is for a fake images. Right? So this is what you see generated image is this uh and um $\hat{y}_{fake}$ is the prediction made by the discriminator for the generated image. And for this the ground truth is zero. So the first term becomes zero and the second term is $1 - \hat{y}$ into this predicted value and the predicted value here is this. So this is what is the overall uh loss function for an generated um images for a discriminator.

[39:16] I hope this is clear. Uh once the terms are clear it is very very simple. It looks very complicated math but it's very very simple. What is the total discriminator loss? Uh recall from the image you have one loss from coming from the real images, one loss coming from the generated image and the total loss is addition of L_{real} and L_{fake} . So this is what is the total loss and just some time back we have seen what is L_{real} this is the formula. What is L_{fake} ? This is the formula. Just substitute back into this. So is it not L_{real} is this and L_{fake} is this? Take out the minus term common when you take the minus term uh out then you have this entire so this is the total discriminator loss. So once you have a

[40:06] discriminator loss you the next step is you will be um computing the gradients. So I hope up till this step it is clear to you. Um yeah I hope uh every math is clear here. What is the next step? It is an back propagation. Now when you look into the diagram back propagation is only up till the discriminator which means generator is not at all getting involved which means those parameters are frozen. What five parameters generator parameters are frozen. So using a chain rule the back propagation is all about uh you know you have to compute the gradients. So using a chain rule if this is the loss then the chain rule says derivative of loss with respect to θ . Why θ ?

Because you have to update the discriminator

[40:59] parameters. Discriminator parameters are θ . So derivative of loss with respect to θ . Derivative of loss fake with respect to θ . So which means now you should compute this and you should compute this. So um uh the gradients has to flow back from where it will flow back the total loss L_D from there you will come to discriminator parameters. So the path is very short here. So uh because of this although you have I_{fake} which is generated images from the generator but during a discriminator training uh I_{real} is frozen. Therefore the gradients are not used to update the generator parameters.

[41:45] You are only going till discriminator and okay now let us see what exactly is these two terms. Once we know these two terms then updating θ becomes easy. Now just recall from that diagram uh how is that you will be first you have to calculate this if you have to calculate uh derivative of loss with respect to θ . What is that path? Is it not that this is the path you have θ uh discriminator discriminator has predicted this from there you have calculated uh I_{real} that is uh loss L_{real} for a real parameters real uh image. If this is the case using chain rule can I write in this way? So L_{real} um derivative of L_{real} with respect to θ can I write it as from L_{real} from loss you will reach to this from here

[42:46] you will be reaching to this parameters θ can we write so derivative of loss with respect to θ is derivative of loss with respect to its predicted value $\hat{y}_{real}$ and $\hat{y}_{real}$ with respect to θ . Now what is $\hat{y}_{real}$? That is a prediction made for a real images. Who makes the prediction discriminator on what it will predict on real images? x . So therefore you have substituted y_{real} $\hat{y}_{real}$ as d_{θ} of x . You have substituted which means each of these values can be easily computed. Right? So I hope this is clear. The path you have to update these you have to get these two values.

[43:32] L_{real} with respect to real images is you will be getting its prediction $\hat{y}_{real}$ from $\hat{y}_{real}$ you can reach to a discriminator parameters. $\hat{y}_{real}$ [snorts] real images also it should able to predict properly and for fake images also it should able to predict. So you are able to reach two parameters and you are updating here. Second, what is for a fake one? For a fake you get I_{fake} um that is loss for a fake images from there you will reach to $\hat{y}_{fake}$ that is the prediction made by the discriminator model on generated images and then you will reach to an discriminator θ . So is it not this is the uh path. So can you able to write using a chain rule? So uh L_{fake} with respect to $\hat{y}_{fake}$ and $\hat{y}_{fake}$ with

[44:25] respect to θ . So what is $\hat{y}_{fake}$? It is uh the prediction made by the discriminator on the generated image. So this is so you have substituted those two values. By substituting those values you have got derivative of uh loss with respect to θ parameters. And for a generated image what is the total loss? The total loss

here is L_{real} plus L_{fake} . So uh this we have obtained and this we have obtained. Just substitute the values add these we are going to get derivative of L with respect to θ .

[45:07] So once we get is it not you can update as θ_{new} is a θ_{old} minus a learning rate derivative of L with respect to θ . So um where this is a learning rate and during this uh discriminator training θ is updated but G is frozen right. So this is all with respect to an discriminator training. Although it looks very complicated but understand the terms thoroughly and then um understand the chain rule you can able to do it very very easily. Now let us see one numerical example uh and we are dealing it with the discriminator.

[45:51] Suppose we have uh L_D of X that is the prediction made by the discriminator on real image is 0.2. The prediction made by the uh discriminator on generated image is 0.2. So is it not very good prediction on real image it is giving 0.9 closer to one. So which means real image is treated as real predicted it as real. So good. Second one 0.2 two is on the generated image what is the discriminator prediction 0.2 2 which is closer to zero which means both these are very good. If both are good which means discriminator is very good ideally discriminator loss should reduce. So let us see that. So what is the loss from this? The loss is very less. So why loss is very less because it is predicted very well. So then take this and then

[46:47] compute the loss. So the loss is once again very less. So compute total loss. What is the total loss? Addition of these two. So overall discriminator loss has reduced here is low. Why it is low? Because for a real image it has assigned high probability. For a predict generated image it has assigned low probability. So therefore it is overall it is a very good uh um model. Now let us take an counter example. Let us take while generated image is assigned low probability. So overall the model has done well. So let us now look at an other side. What is the meaning of this alone? This is a discriminator uh which has predicted on a real image. So the value is 0.2. So prediction made on real image. If the

[47:44] value is low, what is the meaning of it? For a real image, the model is giving less prediction. Which means discriminator is not doing good. So this means a discriminator assigns only 20% probability that uh real image is real. How? For a real image, it has to give more uh probability but it has given less. Okay. Now let us look into this. What is the meaning of this? This is the discriminator has predicted um the prediction on the generated image and it is 0.8 on the generated image discriminator has given high probability. So ideally that value should be closer to zero but here the value is closer to one 0.8 model is not doing good discriminator which means a discriminator has assigned 80% probability that generated image is

[48:36] real. very bad discriminator. What is the total loss? I mean loss on real image. The loss on real image is very high 1.609 and the loss on generated image is

also very high. So once the loss is high then what is the total loss? The total loss is also going very high. when the total loss is high um then during a back propagation what is that information you are conveying to a discriminator you're not doing good if you're not doing good is it not discriminator becomes more active and then try to learn it more carefully that real images should have a probability one and generated images should have a probability closer to zero right so this is all the overall working mechanism and uh I request u um you know um you to

[49:31] take a pen and paper when you are learning this particular um one. So in this particular session, this is very important session where you have seen the basic working mechanism of GAN in a more detailed way and in next coming uh session we will be looking into the GAN objective and then the loss function where we will look into the training procedure and then how GAN will converge and we will look into one of the important called concept called Nash equilibrium. This will be the point of discussion for the next uh uh session.

[50:06] Thank you.